

STAT452

Take-home Mid-term Retake

Comprehensive Solutions

Applied Statistics for Engineers and Scientists II

Term 3, Academic year 2025–2026 · Class 24A02

Conventions. Natural logarithms are used throughout. All tests and intervals use $\alpha_{\text{sig}} = 0.05$. Notation follows the Chapter 1–4 course notes, with exam-specific symbols identified explicitly. Intermediate calculations retain full precision; displayed decimal answers are rounded.

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Notation crosswalk

The following conventions match the Chapter 1–4 notes and the exam.

- In Problems 1–2, p is the number of predictors *excluding* the intercept. Thus X has $p + 1$ columns and $df_E = n - p - 1$. In Problem 1, $p = 2$ and $df_E = 7$. Problem 4 follows its own statement: there, $X \in \mathbb{R}^{n \times p}$ and p is the number of columns of X , equivalently the dimension of β .
- Uppercase symbols such as Y_i , Y , and X denote random variables or model quantities; lowercase y_i , x_i , and v_i denote observed data. Bold symbols denote vectors.
- The notes use $SST = \sum_i (y_i - \bar{y})^2$, $SSE = \sum_i (y_i - \hat{y}_i)^2$, and $SSR = \sum_i (\hat{y}_i - \bar{y})^2$. The exam's RSS is the same quantity as the notes' SSE.
- The significance level is written $\alpha_{\text{sig}} = 0.05$ in this solution. The exam separately uses α for the centered intercept in Problem 1 and for the positive scale parameter in Problem 2.
- In Problem 2, $a = \log \alpha$ is the log-scale intercept and corresponds to β_0 in the Chapter 2 semi-log model.
- Chapter 4 writes $\pi(\mathbf{X}) = P(Y = 1 \mid \mathbf{X})$ and uses threshold τ . Problem 3 uses $\hat{p}(\mathbf{x})$ and c , so $\hat{p}(\mathbf{x}) = \hat{\pi}(\mathbf{x})$ and $c = \tau$.
- $L(\alpha, \beta_1, \beta_2)$ is the least-squares criterion in Problem 1, matching Chapter 1. The exam independently defines $L(c)$ as the threshold cost in Problem 3; the arguments distinguish the two functions.

Quick answer sheet

Item	Main result
1(a)	$\hat{\beta}_0 = 7, \hat{\beta}_1 = 1.5, \hat{\beta}_2 = -1.$
1(b)	$SSR = 35, SSE = 5, R^2 = 0.875, R_{\text{adj}}^2 = 47/56 = 0.8393, \hat{\sigma}^2 = 5/7.$
1(c)	$\widehat{\text{Var}}(\hat{\beta}) = \begin{pmatrix} 29/42 & -5/42 & -1/21 \\ -5/42 & 1/21 & -1/42 \\ -1/21 & -1/42 & 1/21 \end{pmatrix}.$
1(d)	$t = -\sqrt{21} = -4.5826, F = 21 = t^2, p = 0.002536; \text{reject } H_0.$
1(e)	$\hat{y}_0 = 12.5; \text{mean CI } (11.405, 13.595); \text{prediction interval } (10.221, 14.779).$
2(a)	$\hat{a} = 1, \hat{\beta} = 0.5, \hat{\alpha} = e = 2.7183; e^{\hat{\beta}} = 1.6487.$
2(b)	$\hat{\sigma}^2 = 0.025, \text{se}(\hat{\beta}) = 0.03780; \text{CI for } \beta: (0.3951, 0.6049); \text{CI for } e^{\beta}: (1.4845, 1.8311).$
2(c)	Estimated median $= e^4 = 54.5982; \text{mean-corrected estimate} = e^{4.0125} = 55.2849.$
2(d)	Log-scale PI (3.4003, 4.5997); response-scale PI (29.9736, 99.4529).
3(a)–(b)	Conditional odds ratios: 1.0725, 3.0042, 0.2725. $A : \hat{p} = 0.1545 \Rightarrow 0;$ $B : \hat{p} = 0.8520 \Rightarrow 1.$
3(c)–(d)	At $c = 0.50$: all five metrics equal 0.50 and $L(0.50) = 10$. At $c = 0.30$: accuracy 0.50, precision 0.50, recall 0.75, specificity 0.25, $F_1 = 0.60$, and $L(0.30) = 7$. Select $c = 0.30$.
4	For $D = C - (X^T X)^{-1} X^T, \text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = \sigma^2 D D^T \succeq 0.$

1 Problem 1: Multiple linear regression

The model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2),$$

with $n = 10$. Define

$$\mathbf{z}_i = \begin{pmatrix} x_{i1} - 4 \\ x_{i2} - 3 \end{pmatrix}, \quad w_i = y_i - 10,$$

so that $\sum_i \mathbf{z}_i = \mathbf{0}$ and $\sum_i w_i = 0$. The given cross-products are

$$G = \sum_{i=1}^{10} \mathbf{z}_i \mathbf{z}_i^\top = \begin{pmatrix} 20 & 10 \\ 10 & 20 \end{pmatrix}, \quad \mathbf{g} = \sum_{i=1}^{10} \mathbf{z}_i w_i = \begin{pmatrix} 20 \\ -5 \end{pmatrix}.$$

1.1 Part (a): Centered normal equations and estimates

Write $z_{i1} = x_{i1} - 4$ and $z_{i2} = x_{i2} - 3$, so that $\mathbf{z}_i = (z_{i1}, z_{i2})^\top$. The centered intercept specified by the exam is

$$\alpha = \beta_0 + 4\beta_1 + 3\beta_2.$$

Hence

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} = \alpha + \beta_1 z_{i1} + \beta_2 z_{i2}.$$

Because $y_i = \bar{y} + w_i$, the least-squares criterion becomes

$$\begin{aligned} L(\alpha, \beta_1, \beta_2) &= \sum_{i=1}^{10} [y_i - \{\alpha + \beta_1 z_{i1} + \beta_2 z_{i2}\}]^2 \\ &= \sum_{i=1}^{10} [w_i - \beta_1 z_{i1} - \beta_2 z_{i2} - (\alpha - \bar{y})]^2. \end{aligned}$$

Differentiating with respect to α , and using $\sum_i w_i = \sum_i z_{i1} = \sum_i z_{i2} = 0$,

$$\frac{\partial L}{\partial \alpha} = -2 \sum_{i=1}^{10} [w_i - \beta_1 z_{i1} - \beta_2 z_{i2} - (\alpha - \bar{y})] = 20(\alpha - \bar{y}).$$

Thus $\hat{\alpha} = \bar{y} = 10$. At this value,

$$L(10, \beta_1, \beta_2) = \sum_{i=1}^{10} (w_i - \beta_1 z_{i1} - \beta_2 z_{i2})^2.$$

For $j = 1, 2$, the slope normal equations are

$$\frac{\partial L}{\partial \beta_j} = -2 \sum_{i=1}^{10} z_{ij} (w_i - \beta_1 z_{i1} - \beta_2 z_{i2}) = 0.$$

Stacking these two equations gives

$$\begin{pmatrix} \sum_i z_{i1}^2 & \sum_i z_{i1} z_{i2} \\ \sum_i z_{i1} z_{i2} & \sum_i z_{i2}^2 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \sum_i z_{i1} w_i \\ \sum_i z_{i2} w_i \end{pmatrix},$$

which, using the notation supplied by the exam, is

$$G \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \mathbf{g}.$$

Now

$$G^{-1} = \frac{1}{20(20) - 10(10)} \begin{pmatrix} 20 & -10 \\ -10 & 20 \end{pmatrix} = \begin{pmatrix} 1/15 & -1/30 \\ -1/30 & 1/15 \end{pmatrix}.$$

Therefore,

$$\begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = G^{-1} \mathbf{g} = \begin{pmatrix} 1/15 & -1/30 \\ -1/30 & 1/15 \end{pmatrix} \begin{pmatrix} 20 \\ -5 \end{pmatrix} = \begin{pmatrix} 3/2 \\ -1 \end{pmatrix}.$$

Finally,

$$\hat{\beta}_0 = \hat{\alpha} - 4\hat{\beta}_1 - 3\hat{\beta}_2 = 10 - 4(3/2) - 3(-1) = 7.$$

Result.

$$\hat{y} = 7 + 1.5x_1 - x_2.$$

1.2 Part (b): Regression sums of squares and variance estimate

In centered regression, the explained sum of squares is

$$\text{SSR} = \begin{pmatrix} \hat{\beta}_1 & \hat{\beta}_2 \end{pmatrix} \mathbf{g} = \begin{pmatrix} 3/2 & -1 \end{pmatrix} \begin{pmatrix} 20 \\ -5 \end{pmatrix} = 30 + 5 = 35.$$

Since $\text{SST} = 40$,

$$\text{SSE} = \text{SST} - \text{SSR} = 40 - 35 = 5.$$

Thus

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{35}{40} = \boxed{0.875}.$$

There are $p = 2$ predictors and $p + 1 = 3$ fitted coefficients, so the residual degrees of freedom are $df_E = n - p - 1 = 10 - 2 - 1 = 7$. Hence

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE} / (n - p - 1)}{\text{SST} / (n - 1)} = 1 - \frac{5/7}{40/9} = \frac{47}{56} = \boxed{0.839286},$$

$$\hat{\sigma}^2 = \text{MSE} = \frac{\text{SSE}}{n - p - 1} = \boxed{\frac{5}{7} = 0.714286}, \quad \hat{\sigma} = \sqrt{\frac{5}{7}} = \boxed{0.845154}.$$

1.3 Part (c): Covariance under centering and transformation

Let Z be the 10×2 matrix whose i -th row is $\mathbf{z}_i^\top$, and let the centered design matrix be

$$W = \begin{pmatrix} \mathbf{1} & Z \end{pmatrix}.$$

Because every column of Z is centered, $\mathbf{1}^\top Z = \mathbf{0}^\top$, while $Z^\top Z = G$. Therefore

$$W^\top W = \begin{pmatrix} \mathbf{1}^\top \mathbf{1} & \mathbf{1}^\top Z \\ Z^\top \mathbf{1} & Z^\top Z \end{pmatrix} = \begin{pmatrix} 10 & \mathbf{0}^\top \\ \mathbf{0} & G \end{pmatrix}.$$

For $\boldsymbol{\theta} = (\alpha, \beta_1, \beta_2)^\top$, the OLS covariance is

$$\text{Var}(\hat{\boldsymbol{\theta}}) = \sigma^2 (W^\top W)^{-1} = \sigma^2 \begin{pmatrix} 1/10 & \mathbf{0}^\top \\ \mathbf{0} & G^{-1} \end{pmatrix}.$$

Replacing σ^2 by $5/7$ gives the estimated version requested in the question.

To return to the original parameterization, use

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = A \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & -4 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$\begin{aligned} \widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) &= \hat{\sigma}^2 A \begin{pmatrix} 1/10 & 0 & 0 \\ 0 & 1/15 & -1/30 \\ 0 & -1/30 & 1/15 \end{pmatrix} A^\top \\ &= \frac{5}{7} \begin{pmatrix} 29/30 & -1/6 & -1/15 \\ -1/6 & 1/15 & -1/30 \\ -1/15 & -1/30 & 1/15 \end{pmatrix} \\ &= \boxed{\begin{pmatrix} 29/42 & -5/42 & -1/21 \\ -5/42 & 1/21 & -1/42 \\ -1/21 & -1/42 & 1/21 \end{pmatrix}}. \end{aligned}$$

In decimal form, this is

$$\begin{pmatrix} 0.690476 & -0.119048 & -0.047619 \\ -0.119048 & 0.047619 & -0.023810 \\ -0.047619 & -0.023810 & 0.047619 \end{pmatrix}.$$

Notice why centering is useful: $\hat{\alpha}$ is uncorrelated with both slope estimators, even though $\hat{\beta}_0$ generally is not.

1.4 Part (d): Testing $H_0 : \beta_2 = 0$

1.4.1 Coefficient t test

From the (3, 3) entry of the covariance matrix,

$$\text{SE}(\hat{\beta}_2) = \sqrt{\hat{\sigma}^2 (G^{-1})_{22}} = \sqrt{\frac{5}{7} \cdot \frac{1}{15}} = \sqrt{\frac{1}{21}}.$$

Therefore,

$$t = \frac{\hat{\beta}_2 - 0}{\text{SE}(\hat{\beta}_2)} = \frac{-1}{\sqrt{1/21}} = -\sqrt{21} = \boxed{-4.582576}, \quad df = 7.$$

The two-sided p -value is

$$2P(T_7 \geq |t|) = \boxed{0.002536}.$$

1.4.2 Nested-model F test

The reduced model has $\text{RSS}_{\text{red}} = 20$, while the full model has $\text{RSS}_{\text{full}} = \text{SSE} = 5$. There is $q = 1$ restriction, so

$$F = \frac{(\text{RSS}_{\text{red}} - \text{RSS}_{\text{full}})/q}{\text{RSS}_{\text{full}}/(n - p - 1)} = \frac{(20 - 5)/1}{5/7} = \boxed{21}, \quad (df_1, df_2) = (1, 7).$$

Also,

$$t^2 = (-\sqrt{21})^2 = 21 = F.$$

Both methods therefore give the same p -value and the same decision:

$$\boxed{\text{Reject } H_0 \text{ at the 5\% level.}}$$

After controlling for x_1 , the data provide strong evidence that the coefficient of x_2 is nonzero (and its fitted sign is negative).

1.5 Part (e): Mean-response and prediction intervals at $x_0 = (1, 5, 2)^\top$

The centered predictor vector is

$$z_0 = \begin{pmatrix} 5 - 4 \\ 2 - 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The fitted response is

$$\hat{y}_0 = 7 + 1.5(5) - 1(2) = \boxed{12.5}.$$

The leverage is

$$\begin{aligned} h_0 &= \frac{1}{10} + z_0^\top G^{-1} z_0 \\ &= \frac{1}{10} + \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1/15 & -1/30 \\ -1/30 & 1/15 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &= \frac{1}{10} + \frac{1}{5} = \boxed{\frac{3}{10}}. \end{aligned}$$

For the conditional mean, the standard error is

$$\hat{\sigma}\sqrt{h_0} = \sqrt{\frac{5}{7} \cdot \frac{3}{10}} = \sqrt{\frac{3}{14}} = 0.462910.$$

Using $t_{0.975,7} = 2.365$,

$$12.5 \pm 2.365(0.462910) = 12.5 \pm 1.094608,$$

so the 95% confidence interval for $E(Y \mid x_0)$ is

$$\boxed{(11.4054, 13.5946)}.$$

For a new response, the standard error includes the new error term:

$$\hat{\sigma}\sqrt{1+h_0} = \sqrt{\frac{5}{7}\left(1 + \frac{3}{10}\right)} = \sqrt{\frac{13}{14}} = 0.963624.$$

Thus the 95% prediction interval is

$$12.5 \pm 2.365(0.963624) = 12.5 \pm 2.278609 = \boxed{(10.2214, 14.7786)}.$$

The prediction interval is wider because it accounts for both uncertainty in the estimated mean and irreducible variation in a future response.

Calculator verification (Casio fx-580VN X).

1. In `MENU` $\rightarrow$ `Matrix`, define

$$\text{MatA} = \begin{pmatrix} 20 & 10 \\ 10 & 20 \end{pmatrix}, \quad \text{MatB} = \begin{pmatrix} 20 \\ -5 \end{pmatrix}.$$

Evaluate $\text{MatA}^{-1}\text{MatB}$. The display should be $(1.5, -1)^T$.

2. In ordinary Calculate mode, enter $\sqrt{(5 \div 7)(1 \div 15)}$. The display should be 0.2182179; then $-1 \div \text{Ans}$ gives -4.582576 .
3. Enter $(20 - 5) \div (5 \div 7)$. The display should be 21, confirming $F = t^2$.
4. For part (e), enter $12.5 \pm 2.365\sqrt{(5 \div 7)(3 \div 10)}$, then replace $3 \div 10$ inside the square root by $1 + 3 \div 10$. This reproduces the CI and PI endpoints above.

2 Problem 2: Transformation and back-transformation

The response model is

$$Y = \alpha \exp(\beta X) \exp(\varepsilon), \quad \varepsilon \sim N(0, \sigma^2), \quad \alpha > 0.$$

2.1 Part (a): Linearization, estimation, and interpretation

Taking natural logarithms and writing $V = \log Y$ gives

$$V = \log \alpha + \beta X + \varepsilon = a + \beta X + \varepsilon, \quad a = \log \alpha.$$

Thus a is exactly the log-scale intercept denoted by β_0 in the Chapter 2 semi-log model; the symbol a is retained here because it is the notation specified by the exam. This is an ordinary simple linear regression on the log-response scale. The least-squares slope is

$$\hat{\beta} = \frac{S_{xv}}{S_{xx}} = \frac{35/4}{35/2} = \boxed{\frac{1}{2}}.$$

The intercept is

$$\hat{a} = \bar{v} - \hat{\beta} \bar{x} = \frac{9}{4} - \frac{1}{2} \left(\frac{5}{2} \right) = \frac{9}{4} - \frac{5}{4} = \boxed{1}.$$

Therefore,

$$\hat{\alpha} = \exp(\hat{a}) = \boxed{e = 2.718282}.$$

Holding all else fixed, increasing X by one changes the fitted median from $\exp(\hat{a} + \hat{\beta}x)$ to $\exp(\hat{a} + \hat{\beta}(x+1))$. Their ratio is

$$\exp(\hat{\beta}) = e^{1/2} = \boxed{1.648721}.$$

Thus each one-unit increase in X multiplies the conditional median of Y by about 1.6487, corresponding to an estimated 64.87% increase.

2.2 Part (b): Error variance and confidence interval for the effect

There are $n - 2 = 4$ residual degrees of freedom. Hence

$$\hat{\sigma}^2 = \text{MSE}_{\log} = \frac{\text{SSE}_{\log}}{n - 2} = \frac{0.10}{4} = \boxed{0.025}, \quad \hat{\sigma} = \sqrt{0.025} = 0.158114.$$

The slope standard error is

$$\text{se}(\hat{\beta}) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} = \sqrt{\frac{0.025}{35/2}} = \sqrt{\frac{1}{700}} = \boxed{0.037796}.$$

Using $t_{0.975,4} = 2.776$,

$$\begin{aligned} \hat{\beta} \pm t_{0.975,4} \text{se}(\hat{\beta}) &= 0.5 \pm 2.776(0.037796) \\ &= 0.5 \pm 0.104923. \end{aligned}$$

Thus the 95% confidence interval for β is

$$\boxed{(0.395077, 0.604923)}.$$

Since the exponential function is strictly increasing, exponentiating the two endpoints gives the 95% interval for the one-unit multiplicative effect:

$$(e^{0.395077}, e^{0.604923}) = (1.484499, 1.831111).$$

In words, a one-unit increase in X is associated with a multiplication of the conditional median by between about 1.48 and 1.83.

2.3 Part (c): Conditional median and lognormal mean correction

At $x_0 = 6$,

$$\hat{v}_0 = \hat{a} + \hat{\beta}x_0 = 1 + \frac{1}{2}(6) = 4.$$

Because a normal error has median zero, the fitted conditional median is

$$\widehat{\text{Med}}(Y \mid X = 6) = \exp(\hat{v}_0) = e^4 = 54.598150.$$

The stated lognormal mean correction gives

$$\hat{E}(Y \mid X = 6) = \exp\left(\hat{v}_0 + \frac{\hat{\sigma}^2}{2}\right) = \exp\left(4 + \frac{0.025}{2}\right) = e^{4.0125} = 55.284913.$$

They differ because, although $\text{Med}(e^\varepsilon) = 1$,

$$E(e^\varepsilon) = e^{\sigma^2/2} > 1.$$

Exponentiation makes the response distribution right-skewed; large positive errors increase the mean more than equally sized negative errors decrease it. The mean is therefore larger than the median.

2.4 Part (d): Prediction interval at $x_0 = 6$

The leverage for predicting on the log scale is

$$\begin{aligned} h_0 &= \frac{1}{6} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \\ &= \frac{1}{6} + \frac{(6 - 5/2)^2}{35/2} = \frac{1}{6} + \frac{(7/2)^2}{35/2} \\ &= \frac{1}{6} + \frac{7}{10} = \frac{13}{15} = 0.866667. \end{aligned}$$

The prediction standard error on the log scale is

$$\hat{\sigma}\sqrt{1+h_0} = \sqrt{0.025\left(1 + \frac{13}{15}\right)} = \sqrt{\frac{7}{150}} = 0.216025.$$

Thus

$$\hat{v}_0 \pm t_{0.975,4}\hat{\sigma}\sqrt{1+h_0} = 4 \pm 2.776(0.216025) = 4 \pm 0.599685.$$

The 95% prediction interval for $V_0 = \log Y_0$ is

$$(3.400315, 4.599685).$$

Exponentiating both endpoints,

$$(e^{3.400315}, e^{4.599685}) = (29.973550, 99.452943)$$

is the 95% prediction interval for a future response Y_0 . It is asymmetric about 54.598 on the original scale, as expected after exponentiation.

Calculator verification (Casio fx-580VN X).

1. In Calculate mode, enter $(35 \div 4) \div (35 \div 2)$; the display should be 0.5. Then enter $9 \div 4 - 0.5(5 \div 2)$; the display should be 1.
 2. Enter $\sqrt{0.025 \div (35 \div 2)}$; the display should be 0.03779645. The two expressions $0.5 - 2.776 \text{ Ans}$ and $0.5 + 2.776 \text{ Ans}$ give the slope interval endpoints.
 3. Use the e^x function on those endpoints to obtain 1.484499 and 1.831111.
 4. Enter e^4 and $e^{4+0.025/2}$ to obtain 54.59815 and 55.28491. For the prediction interval, evaluate $e^{4 \pm 2.776 \sqrt{0.025(1+13/15)}}$.
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3 Problem 3: Logistic regression evaluation

The fitted logit is

$$\log \left(\frac{\hat{p}(\mathbf{x})}{1 - \hat{p}(\mathbf{x})} \right) = -5 + 0.07x_1 + 1.10x_2 - 1.30x_3.$$

In the notation of Chapter 4, $\pi(\mathbf{X}) = P(Y = 1 \mid \mathbf{X})$, so $\hat{p}(\mathbf{x}) = \hat{\pi}(\mathbf{x})$. The exam's cutoff c is the same classification threshold denoted by τ in the notes.

Following Chapter 4 directly, the fitted probability is obtained by applying its logistic function to the fitted log-odds:

$$\hat{p}(\mathbf{x}) = \frac{1}{1 + \exp[-\{-5 + 0.07x_1 + 1.10x_2 - 1.30x_3\}]}$$

3.1 Part (a): Conditional odds-ratio interpretations

Each interpretation holds with the other predictors fixed.

- A one-unit increase in temperature x_1 multiplies the alarm odds by

$$e^{0.07} = \boxed{1.072508},$$

an increase of approximately 7.25%.

- A one-unit increase in vibration x_2 multiplies the alarm odds by

$$e^{1.10} = \boxed{3.004166},$$

so the odds are approximately tripled.

- Changing from no preventive maintenance ($x_3 = 0$) to preventive maintenance ($x_3 = 1$) multiplies the alarm odds by

$$e^{-1.30} = \boxed{0.272532}.$$

Thus preventive maintenance is associated with a $1 - 0.272532 = 72.75\%$ reduction in the conditional odds of an alarm. Equivalently, the odds without maintenance are $e^{1.30} = 3.6693$ times those with maintenance.

3.2 Part (b): Probabilities and classification at threshold 0.5

For $A = (50, 1, 1)$,

$$\log \left(\frac{\hat{p}_A}{1 - \hat{p}_A} \right) = -5 + 0.07(50) + 1.10(1) - 1.30(1) = -1.70,$$

and therefore

$$\hat{p}_A = \frac{1}{1 + e^{1.70}} = \boxed{0.154465}.$$

Since $\hat{p}_A < 0.5$, the fitted class is $\boxed{0}$.

For $B = (65, 2, 0)$,

$$\log \left(\frac{\hat{p}_B}{1 - \hat{p}_B} \right) = -5 + 0.07(65) + 1.10(2) - 1.30(0) = 1.75,$$

so

$$\hat{p}_B = \frac{1}{1 + e^{-1.75}} = \boxed{0.851953}.$$

Since $\hat{p}_B \geq 0.5$, the fitted class is $\boxed{1}$.

3.3 Part (c): Confusion matrices and metrics

The metric definitions are

$$\begin{aligned} \text{accuracy} &= \frac{TP + TN}{TP + FP + TN + FN}, & \text{precision} &= \frac{TP}{TP + FP}, \\ \text{recall} &= \frac{TP}{TP + FN}, & \text{specificity} &= \frac{TN}{TN + FP}, \\ F_1 &= \frac{2(\text{precision})(\text{recall})}{\text{precision} + \text{recall}} = \frac{2TP}{2TP + FP + FN}. \end{aligned}$$

3.3.1 Threshold $c = 0.50$

The predicted labels are

$$(\hat{Y}_1, \dots, \hat{Y}_8) = (1, 1, 1, 1, 0, 0, 0, 0).$$

Comparing these to $(1, 0, 1, 0, 1, 0, 1, 0)$ gives

$$TP = 2, \quad FP = 2, \quad TN = 2, \quad FN = 2.$$

Hence

$$\begin{aligned} \text{accuracy} &= \frac{2+2}{8} = 0.50, & \text{precision} &= \frac{2}{2+2} = 0.50, \\ \text{recall} &= \frac{2}{2+2} = 0.50, & \text{specificity} &= \frac{2}{2+2} = 0.50, \\ F_1 &= \frac{2(2)}{2(2)+2+2} = \boxed{0.50}. \end{aligned}$$

3.3.2 Threshold $c = 0.30$

The predicted labels are

$$(\hat{Y}_1, \dots, \hat{Y}_8) = (1, 1, 1, 1, 1, 1, 0, 0).$$

Therefore,

$$TP = 3, \quad FP = 3, \quad TN = 1, \quad FN = 1.$$

The metrics are

$$\begin{aligned} \text{accuracy} &= \frac{3+1}{8} = \boxed{0.50}, & \text{precision} &= \frac{3}{3+3} = \boxed{0.50}, \\ \text{recall} &= \frac{3}{3+1} = \boxed{0.75}, & \text{specificity} &= \frac{1}{1+3} = \boxed{0.25}, \\ F_1 &= \frac{2(3)}{2(3)+3+1} = \boxed{0.60}. \end{aligned}$$

c	TP	FP	TN	FN	Accuracy	Precision	Recall
0.50	2	2	2	2	0.50	0.50	0.50
0.30	3	3	1	1	0.50	0.50	0.75

c	Specificity	F_1
0.50	0.50	0.50
0.30	0.25	0.60

Lowering the threshold finds one additional true alarm (recall rises), but it also creates one additional false alarm (specificity falls). Accuracy and precision happen to remain unchanged in this test set.

3.4 Part (d): Cost-sensitive threshold choice

Using $L(c) = 4FN(c) + FP(c)$,

$$L(0.50) = 4(2) + 2 = \boxed{10}, \quad L(0.30) = 4(1) + 3 = \boxed{7}.$$

Therefore,

$$\boxed{\text{Select } c = 0.30.}$$

Although both thresholds have the same accuracy, the lower threshold avoids one costly false negative. Its one extra false positive costs only 1, so the total empirical cost decreases by 3.

Calculator verification (Casio fx-580VN X). In Calculate mode:

1. For case *A*, enter $-5 + 0.07(50) + 1.10(1) - 1.30(1)$; the display is -1.7 . Then enter $1 \div (1 + e^{-\text{Ans}})$; the display is 0.1544653.
 2. Repeat with *B*; the linear predictor is 1.75 and the probability is 0.8519528.
 3. For each threshold, write 1 beside every probability at least as large as c , then tally the four cells. The two tallies should be $(TP, FP, TN, FN) = (2, 2, 2, 2)$ and $(3, 3, 1, 1)$.
 4. Enter $4(2) + 2$ and $4(1) + 3$; the displays 10 and 7 verify the cost comparison.
-

4 Problem 4: Gauss–Markov theorem

Assume

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I_n,$$

where $X \in \mathbb{R}^{n \times p}$ is deterministic and $\text{rank}(X) = p$. Here p follows the notation of Problem 4: it is the number of columns of X and the dimension of $\boldsymbol{\beta}$, not the predictor count convention used in Problem 1. No distributional assumption beyond these first two moments is needed.

4.1 Step 1: Error and covariance of OLS

Let

$$A = (X^\top X)^{-1} X^\top, \quad \hat{\boldsymbol{\beta}} = AY.$$

Full column rank guarantees that $X^\top X$ is invertible, and

$$AX = (X^\top X)^{-1} X^\top X = I_p.$$

Substituting $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$,

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (X^\top X)^{-1} X^\top (X\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}. \end{aligned}$$

Therefore,

$$\boxed{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}}.$$

Its covariance is

$$\begin{aligned} \text{Var}(\hat{\boldsymbol{\beta}}) &= A \text{Var}(\boldsymbol{\varepsilon}) A^\top \\ &= \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} \\ &= \boxed{\sigma^2 (X^\top X)^{-1}}. \end{aligned}$$

4.2 Step 2: What unbiasedness of a general linear estimator implies

Let $\tilde{\boldsymbol{\beta}} = C\mathbf{Y}$, where $C \in \mathbb{R}^{p \times n}$ is deterministic. Then

$$\mathbb{E}(\tilde{\boldsymbol{\beta}}) = C \mathbb{E}(\mathbf{Y}) = CX\boldsymbol{\beta}.$$

Unbiasedness for every $\boldsymbol{\beta} \in \mathbb{R}^p$ requires

$$CX\boldsymbol{\beta} = \boldsymbol{\beta} \quad \text{for every } \boldsymbol{\beta}.$$

Equivalently, $(CX - I_p)\boldsymbol{\beta} = \mathbf{0}$ for every $\boldsymbol{\beta}$, which can hold only when

$$\boxed{CX = I_p}.$$

4.3 Step 3: Decomposition and vanishing cross terms

Define

$$D = C - A = C - (X^T X)^{-1} X^T.$$

Using $CX = I_p$ and $AX = I_p$,

$$DX = CX - AX = I_p - I_p = \boxed{0}.$$

Since $C = A + D$,

$$\begin{aligned} \text{Var}(\tilde{\beta}) &= C \text{Var}(\mathbf{Y}) C^T \\ &= \sigma^2 (A + D)(A + D)^T \\ &= \sigma^2 (AA^T + AD^T + DA^T + DD^T). \end{aligned}$$

The cross terms are zero. Indeed,

$$AD^T = (X^T X)^{-1} X^T D^T = (X^T X)^{-1} (DX)^T = 0,$$

and, equivalently by transposition,

$$DA^T = DX(X^T X)^{-1} = 0.$$

Moreover, $\text{Var}(\hat{\beta}) = \sigma^2 AA^T$. Consequently,

$$\boxed{\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = \sigma^2 DD^T}.$$

For every $\mathbf{u} \in \mathbb{R}^p$,

$$\mathbf{u}^T (\sigma^2 DD^T) \mathbf{u} = \sigma^2 (D^T \mathbf{u})^T (D^T \mathbf{u}) = \sigma^2 \|D^T \mathbf{u}\|^2 \geq 0.$$

Hence $\sigma^2 DD^T$ is positive semidefinite, proving

$$\boxed{\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) \succeq 0}.$$

Result. Among all linear estimators that are unbiased for every β , OLS has the smallest covariance matrix in the positive-semidefinite (Loewner) order. The proof used only linearity, unbiasedness, and $\text{Var}(\varepsilon) = \sigma^2 I_n$; it did not use Gaussian errors.

Calculator verification (Casio fx-580VN X). No numerical calculator step is needed for this proof. A useful dimension check is

$$C, D, A \in \mathbb{R}^{p \times n}, \quad X \in \mathbb{R}^{n \times p}, \quad DX \in \mathbb{R}^{p \times p}, \quad DD^T \in \mathbb{R}^{p \times p}.$$

These dimensions confirm that every matrix product in the covariance decomposition is conformable.

A Formula reference aligned with Chapters 1–4

This appendix collects the formulas used in the solutions with the notation of the course notes. In the Chapter 1–3 regression formulas, n is the sample size and p is the number of predictors excluding the intercept. Problem 4 retains the exam's separate convention $X \in \mathbb{R}^{n \times p}$, where $p = \dim(\beta)$. Formulas explicitly supplied by the exam are labelled as such rather than attributed to the notes. To avoid the exam's reuse of α , the significance level is denoted by α_{sig} .

A.1 Chapter 1: Multiple linear regression

Model, least squares, and estimation

$$\mathbf{Y} = X\beta + \varepsilon, \quad \varepsilon \sim N(\mathbf{0}, \sigma^2 I_n), \quad L(\beta) = (\mathbf{Y} - X\beta)^\top (\mathbf{Y} - X\beta).$$

The normal equations and least-squares estimator are

$$X^\top X \hat{\beta} = X^\top \mathbf{Y}, \quad \hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{Y}.$$

With p predictors, the error degrees of freedom and variance estimate are

$$df_E = n - p - 1, \quad \hat{\sigma}^2 = \text{MSE} = \frac{\text{SSE}}{n - p - 1}, \quad \text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Covariance and coefficient inference

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}, \quad \text{SE}(\hat{\beta}_j) = \sqrt{\text{MSE} [(X^\top X)^{-1}]_{jj}}.$$

For $H_0 : \beta_j = \beta_{j,0}$,

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)}, \quad t \sim t_{n-p-1} \quad \text{under } H_0,$$

and the $100(1 - \alpha_{\text{sig}})\%$ confidence interval is

$$\hat{\beta}_j \pm t_{1-\alpha_{\text{sig}}/2, n-p-1} \text{SE}(\hat{\beta}_j).$$

Sums of squares and goodness of fit

$$\text{SST} = \sum_i (y_i - \bar{y})^2, \quad \text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \sum_i (y_i - \hat{y}_i)^2, \quad \text{SST} = \text{SSR} + \text{SSE}.$$

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}, \quad R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}.$$

Nested-model partial F test

If $q = df_{E,\text{red}} - df_{E,\text{full}}$ restrictions separate the reduced and full models, then

$$F = \frac{(\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}})/q}{\text{MSE}_{\text{full}}} \sim F_{q, df_{E,\text{full}}} \quad \text{under } H_0.$$

The exam writes these residual sums of squares as RSS. For one restriction, the coefficient test and nested-model test satisfy $F = t^2$.

Prediction formulas supplied by the exam

For a centered predictor vector $\mathbf{z}_0$,

$$h_0 = \frac{1}{n} + \mathbf{z}_0^T G^{-1} \mathbf{z}_0.$$

The mean-response confidence interval and new-response prediction interval are

$$\hat{y}_0 \pm t_{1-\alpha_{\text{sig}}/2, n-p-1} \hat{\sigma} \sqrt{h_0}, \quad \hat{y}_0 \pm t_{1-\alpha_{\text{sig}}/2, n-p-1} \hat{\sigma} \sqrt{1 + h_0}.$$

A.2 Chapter 2: Transformation of variables

For the semi-log model in Problem 2,

$$Y = \alpha e^{\beta X} e^{\varepsilon} \implies V = \log Y = a + \beta X + \varepsilon, \quad a = \log \alpha.$$

This is the Chapter 2 formula $\log Y = \beta_0 + \beta_1 X$, with $a \equiv \beta_0$ and $\beta \equiv \beta_1$. The simple-regression estimators are

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}, \quad \hat{a} = \bar{v} - \hat{\beta} \bar{x}, \quad \hat{\alpha} = e^{\hat{a}}.$$

A one-unit increase in X multiplies the conditional median of Y by

$$e^{\hat{\beta}}.$$

The slope standard error and interval are

$$\text{se}(\hat{\beta}) = \sqrt{\frac{\text{MSE}_{\log}}{S_{xx}}}, \quad \hat{\beta} \pm t_{1-\alpha_{\text{sig}}/2, n-2} \text{se}(\hat{\beta}).$$

Exponentiating the endpoints gives an interval for the multiplicative effect e^{β} .

For the lognormal model, the following back-transformations are supplied by the exam:

$$\widehat{\text{Med}}(Y \mid X = x_0) = e^{\hat{a} + \hat{\beta} x_0}, \quad \widehat{\text{E}}(Y \mid X = x_0) = e^{\hat{a} + \hat{\beta} x_0 + \hat{\sigma}^2/2}.$$

For a future log-response,

$$h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}, \quad \hat{v}_0 \pm t_{1-\alpha_{\text{sig}}/2, n-2} \hat{\sigma} \sqrt{1 + h_0}.$$

Exponentiating both endpoints gives the prediction interval for Y_0 .

A.3 Chapter 3: Relevance to this retake

The retake does not ask for variable selection, cross-validation, ridge, or LASSO calculations. The Chapter 3 Wald statistic

$$t = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$$

is the same coefficient statistic used in Problem 1(d). Because Problem 1 is an ordinary least-squares model with normally distributed errors, the exact finite-sample reference distribution is t_{n-p-1} , as stated in Chapter 1.

A.4 Chapter 4: Logistic regression and classification

The notes define

$$\pi(\mathbf{X}) = P(Y = 1 \mid \mathbf{X}), \quad \text{logit}\{\pi(\mathbf{X})\} = \log\left(\frac{\pi(\mathbf{X})}{1 - \pi(\mathbf{X})}\right) = \mathbf{X}^\top \boldsymbol{\beta},$$

and therefore

$$\pi(\mathbf{X}) = \frac{e^{\mathbf{X}^\top \boldsymbol{\beta}}}{1 + e^{\mathbf{X}^\top \boldsymbol{\beta}}} = \frac{1}{1 + e^{-\mathbf{X}^\top \boldsymbol{\beta}}}.$$

The conditional odds and one-unit odds ratio are

$$\text{Odds}(Y = 1 \mid \mathbf{X}) = \frac{\pi}{1 - \pi}, \quad \text{OR}_j = e^{\beta_j}.$$

Using the notes' threshold τ (called c in the exam),

$$\hat{Y} = \mathbf{1}\{\hat{\pi}(\mathbf{X}) \geq \tau\} = \mathbf{1}\{\hat{p}(\mathbf{x}) \geq c\}.$$

The classification metrics are

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN}, & \text{Precision} &= \frac{TP}{TP + FP}, \\ \text{Recall} = \text{TPR} &= \frac{TP}{TP + FN}, & \text{Specificity} &= \frac{TN}{TN + FP} = 1 - \text{FPR}, \\ F_1 &= \frac{2(\text{Precision})(\text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}. \end{aligned}$$

The cost $L(c) = 4FN(c) + FP(c)$ is problem-specific and is supplied directly by the exam.

A.5 Formula-source checklist

Exam part	Formula used	Course source
1(a)	Least-squares criterion, normal equations, and OLS estimator	Chapter 1 notes, p. 2; centered form derived in the solution
1(b)	SST–SSR–SSE decomposition, R^2 , adjusted R^2 , and MSE	Chapter 1 notes, pp. 3–4
1(c)	$\text{Var}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$	Chapter 1 notes, p. 4; reparameterization derived in the solution
1(d)	Coefficient t test and partial F test	Chapter 1 notes, pp. 4–5
1(e)	Mean-response CI and prediction interval	Formulas supplied by the exam
2(a)	Semi-log linearization and e^β interpretation	Chapter 2 notes, p. 5
2(b)	Regression MSE, slope SE, and t interval	Chapter 1 notes, pp. 3–4; quantities supplied by the exam
2(c)–(d)	Lognormal mean correction and log-scale prediction interval	Formulas supplied by the exam; back-transformation principle in Chapter 2 notes, pp. 6–7
3(a)–(b)	Logit, logistic probability, and odds ratios	Chapter 4 notes, pp. 1–3
3(c)	Threshold rule, confusion matrix, and classification metrics	Chapter 4 notes, p. 6
3(d)	Weighted empirical cost	Formula supplied by the exam
4	OLS covariance and unbiasedness; Gauss–Markov covariance difference	Chapter 1 notes, pp. 2 and 4; proof derived in the solution